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✓ PAS Home Assignment 1

Aim

Collect academic data of students, including attendance, study hours, assignment marks, internal assessment scores, and final examination marks, to understand the factors influencing academic performance. Using Python and appropriate libraries such as Pandas, NumPy, and Matplotlib, import the dataset, perform data cleaning and preprocessing, classify the variables into numerical and categorical types, compute descriptive statistical measures, explore the dataset using suitable visualizations, analyze the correlation among variables, and prepare a concise analytical report summarizing the important patterns and insights obtained from the data.

I recommend to check the notebook from Github to get the latest version: [Github](#)

✓ Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

✓ Loading the Dataset

```
df = pd.read_csv('PAS-HA_1-Probability_Statistics_Datasets.csv')
df.head()
```

	Student_ID	Attendance_%	Study_Hours	Assignment_Marks	Internal_Marks	Final_Exam_Marks
0	1	95	1.8	87	70	73
1	2	69	2.0	46	46	54
2	3	57	1.2	53	64	60
3	4	93	1.2	52	88	69
4	5	83	5.1	91	35	65

✓ Understanding the Dataset

```
df.shape
```

```
(100, 6)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Student_ID            100 non-null   int64  
1   Attendance_%          100 non-null   int64  
2   Study_Hours           100 non-null   float64 
3   Assignment_Marks      100 non-null   int64  
4   Internal_Marks        100 non-null   int64  
5   Final_Exam_Marks     100 non-null   int64  
dtypes: float64(1), int64(5)
memory usage: 4.8 KB
```

```
df.columns
```

```
Index(['Student_ID', 'Attendance_%', 'Study_Hours', 'Assignment_Marks',
       'Internal_Marks', 'Final_Exam_Marks'],
      dtype='str')
```

✓ Data Cleaning and Preprocessing

```
df.isnull().sum()
```

```
Student_ID      0
Attendance_%    0
Study_Hours     0
Assignment_Marks 0
Internal_Marks  0
Final_Exam_Marks 0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.dtypes
```

```
Student_ID      int64
Attendance_%    int64
Study_Hours     float64
Assignment_Marks int64
Internal_Marks  int64
Final_Exam_Marks int64
dtype: object
```

The dataset has no missing values and no duplicate rows. All columns already have the correct data types, so no cleaning was required.

Classification of Variables

- **Student_ID** - identifier, not used in statistical analysis.
- **Attendance_%** - numerical (quantitative, continuous)
- **Study_Hours** - numerical (quantitative, continuous)
- **Assignment_Marks** - numerical (quantitative, discrete)
- **Internal_Marks** - numerical (quantitative, discrete)
- **Final_Exam_Marks** - numerical (quantitative, discrete)

The provided dataset contains no categorical variable suitable for analysis.

```
num_cols = ['Attendance_%', 'Study_Hours', 'Assignment_Marks', 'Internal_Marks', 'Final_Exam_Marks']
df_num = df[num_cols]
```

Descriptive Statistics

```
df_num.describe()
```

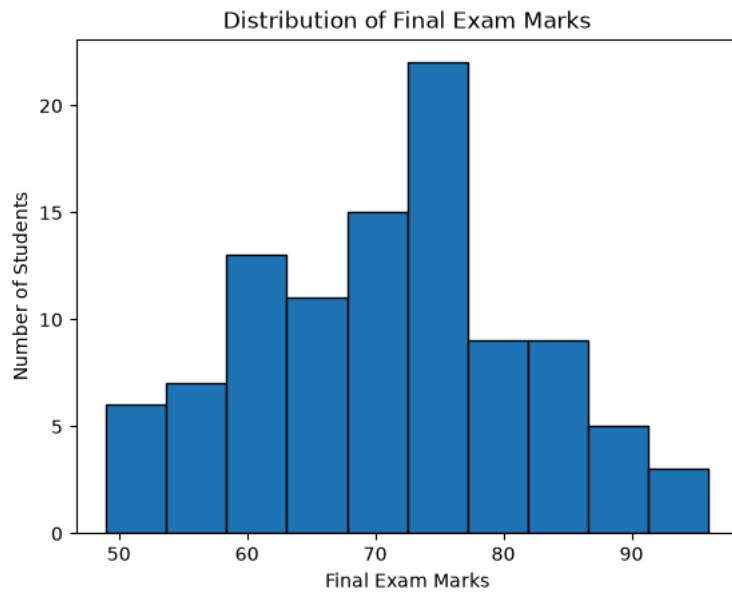
	Attendance_%	Study_Hours	Assignment_Marks	Internal_Marks	Final_Exam_Marks
count	100.00000	100.000000	100.000000	100.000000	100.000000
mean	77.38000	4.270000	74.340000	66.870000	71.140000
std	13.46209	2.146479	16.827839	18.712959	10.571565
min	55.00000	1.100000	42.000000	35.000000	49.000000
25%	65.00000	2.275000	58.750000	49.000000	63.000000
50%	78.50000	4.050000	76.000000	66.500000	72.000000
75%	89.00000	6.025000	88.500000	81.000000	78.000000
max	100.00000	7.800000	99.000000	100.000000	96.000000

```
df_num.var()
```

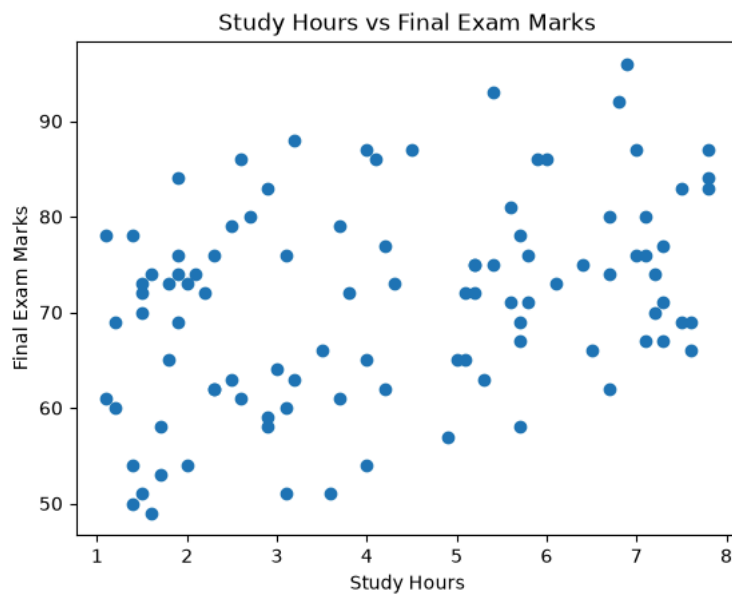
```
Attendance_%    181.227879
Study_Hours     4.607374
Assignment_Marks 283.176162
Internal_Marks  350.174848
Final_Exam_Marks 111.757980
dtype: float64
```

Data Visualization

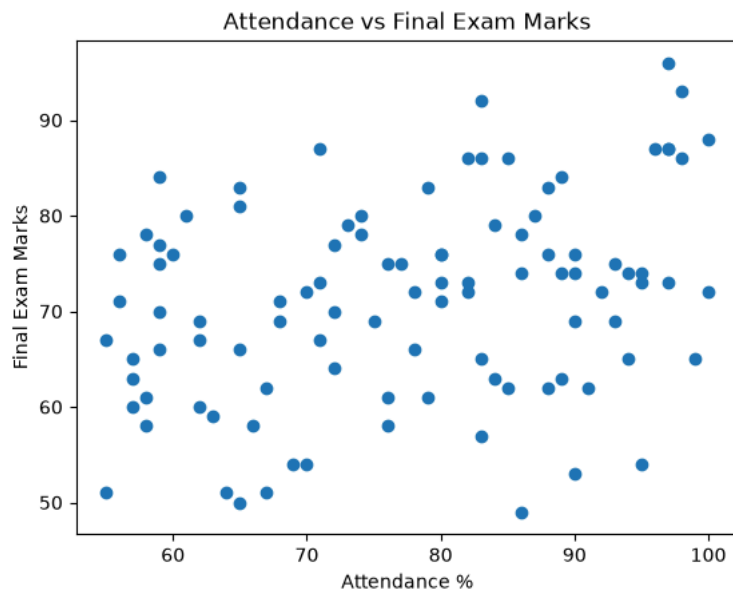
```
plt.hist(df['Final_Exam_Marks'], bins=10, edgecolor='black')  
plt.xlabel('Final Exam Marks')  
plt.ylabel('Number of Students')  
plt.title('Distribution of Final Exam Marks')  
plt.show()
```



```
plt.scatter(df['Study_Hours'], df['Final_Exam_Marks'])  
plt.xlabel('Study Hours')  
plt.ylabel('Final Exam Marks')  
plt.title('Study Hours vs Final Exam Marks')  
plt.show()
```



```
plt.scatter(df['Attendance_%'], df['Final_Exam_Marks'])  
plt.xlabel('Attendance %')  
plt.ylabel('Final Exam Marks')  
plt.title('Attendance vs Final Exam Marks')  
plt.show()
```



Correlation Analysis

```
corr = df_num.corr()
corr
```

	Attendance_%	Study_Hours	Assignment_Marks	Internal_Marks	Final_Exam_Marks
Attendance_%	1.000000	-0.111007	0.050968	0.094907	0.316744
Study_Hours	-0.111007	1.000000	0.044050	-0.116556	0.367518
Assignment_Marks	0.050968	0.044050	1.000000	-0.043226	0.394352
Internal_Marks	0.094907	-0.116556	-0.043226	1.000000	0.598928
Final_Exam_Marks	0.316744	0.367518	0.394352	0.598928	1.000000

```
plt.matshow(corr, cmap='coolwarm')
plt.xticks(range(len(num_cols)), num_cols, rotation=90)
plt.yticks(range(len(num_cols)), num_cols)
plt.colorbar()
plt.title('Correlation Matrix', pad=60)
plt.show()
```

```
corr['Final_Exam_Marks']
```

```
Attendance_%    0.316744
Study_Hours     0.367518
Assignment_Marks 0.394352
Internal_Marks   0.598928
Final_Exam_Marks 1.000000
Name: Final_Exam_Marks, dtype: float64
```

I recommend to check the notebook from Github to get the latest version: [Github](#)

Observations

- Internal Marks has the strongest positive correlation with Final_Exam_Marks (about 0.60), which is a moderate relationship.
- Assignment_Marks (about 0.39) and Study_Hours (about 0.37) also show weak-to-moderate positive correlation with Final_Exam_Marks.
- Attendance_% has the weakest positive correlation with Final_Exam_Marks (about 0.32) among the four variables.
- Final_Exam_Marks ranges from 49 to 96, with a mean of about 71.1 and a standard deviation of about 10.5.
- All correlations are positive, meaning higher values of these variables are generally associated with higher final exam marks, but none of these relationships are very strong, so other factors are also affecting performance.

Conclusion

The dataset of 100 students was clean, with no missing values or duplicates. All five academic variables are numerical, and there are no categorical variables in this dataset. Among the variables studied, Internal_Marks showed the highest positive correlation with Final_Exam_Marks, while Attendance_% showed the lowest. Overall, the analysis shows a positive but moderate relationship between the given academic factors and final exam performance.

